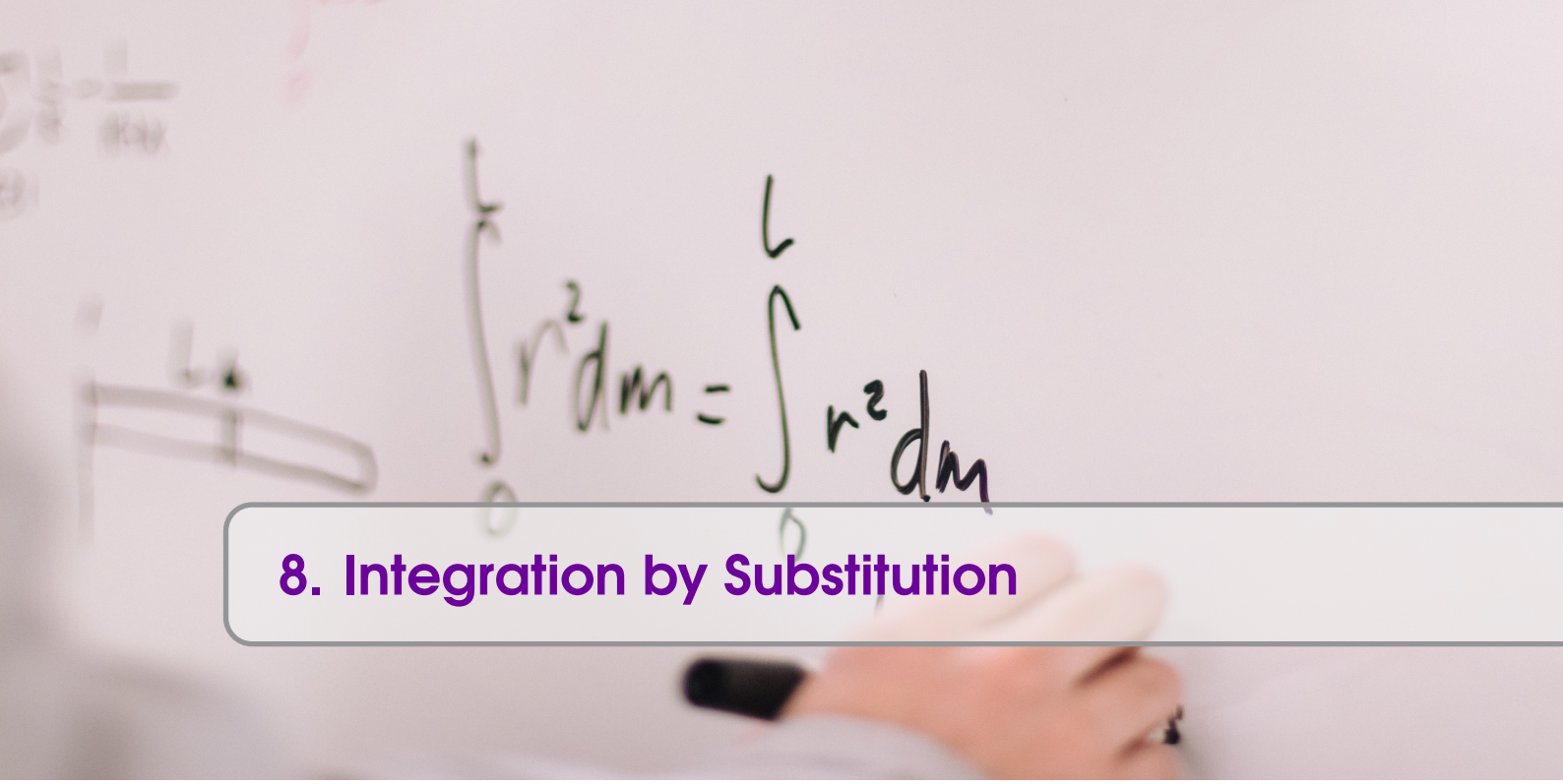


8. Integration by Substitution

8.1 Overview

In Chapters 6 and 7, we introduced rules that allow us to directly integrate certain basic functions if we can recognize their forms. These can be found in this table:

Differential equation $\frac{dy}{dx}$	Integral, y
constant	$constant \times x + d$
x	$\frac{x^2}{2} + d$
$3x^2$	$x^3 + d$
x^3	$\frac{x^4}{4} + d$
$\vdots$	$\vdots$
x^n	$\frac{x^{n+1}}{n+1} + d$
ax^n	$\frac{ax^{n+1}}{n+1} + d$
ke^{kx}	$e^{kx} + d$
$\frac{1}{x}$	$ln(x) + d$
$cos(x)$	$sin(x) + d$
$sin(x)$	$-cos(x) + d$

There may be a need to rearrange the equation, often by using techniques like separation of variables (e.g., rearranging N and t) before we can use them, but they are there for us to use. Furthermore, like differentiation for a sum of functions, the **integral of a sum** is equal to the **sum of the integrals** of the individual functions.

$$\frac{dy}{dx} = u(x) + v(x) \quad \Longrightarrow \quad y = \int u(x)dx + \int v(x)dx$$

In some cases, we cant directly integrate by using these rules, as we are faced with more complicated combinations of functions such as **functions of functions**. For example, what happens if we want to integrate

$$y = \int \frac{dy}{dx} = \int (3x+2)^4 dx$$

This is where **integration by substitution** comes in.

8.2 Integration by substitution

Integration by substitution is essentially the reverse process of the **chain rule** in differentiation. Let's return to the example where we want to evaluate

$$y = \int (3x+2)^4 dx$$

At first glance, it may appear that this resembles the derivative of some function, allowing us to integrate directly. What if we replace the expression in the brackets with another variable such that we can integrate using the rules we already know? Making a **substitution** can often simplify things and enables us to perform what seems to be an otherwise difficult integration.

Lets use $u = 3x+2$ so the integral becomes

$$y = \int u^4 dx$$

Now, the integral is expressed in terms of u, but we need to address the dx. This is asking us to integrate u with respect to x. Since dx relates to x, we need to rewrite it in terms of u. This step ensures that the integral is entirely in terms of the new variable. When the integral contains a factor like $ax+b$, substituting $u = ax+b$ often simplifies the problem. Along with the substitution, we must also determine dx in terms of u to complete the process.

Let's show how **integration by substitution** works in full:

1. Write out the integral in its simplest form:

$$y = \int (3x+2)^4 dx$$

2. Find a new variable (u) as a function of the old variable (x) as a substitution:

$$u = 3x+2$$

3. Rearrange to get an expression for the old variable (x) in terms of the new variable (u):

$$x = \frac{1}{3}(u-2)$$

4. Differentiate with respect to the new variable u:

$$\frac{dx}{du} = \frac{1}{3}$$

5. Get an expression for dx:

$$dx = \frac{du}{3}$$

6. Make the substitution in the original integral

$$y = \int (3x+2)^4 dx \implies y = \int u^4 \frac{du}{3} \implies y = \frac{1}{3}\int u^4 \, du$$

7. Integrate as usual:

$$y = \frac{1}{3}\int u^4 \, du \implies \frac{1}{15}u^5 + d$$

8. Substitute the original expression back in to get the final expression:

$$y = \frac{1}{15}(3x+2)^5 + d$$

Let's see some more examples of this.

Example 8.1: Lets evaluate the integral

$$y = \int (x+4)^5 \, dx$$

Let's use $u = x+4$ as a substitution. Lets rearrange for x, differentiate in terms of u and find an expression for dx:

$$u = x+4 \implies x = u-4 \implies \frac{dx}{du} = 1 \implies dx = du$$

Use these to make the substitution:

$$\int (x+4)^5 \, dx \implies \int u^5 du$$

Now we can evaluate using the basic laws of integration:

$$\int u^5 du = \frac{1}{6}u^6 + c$$

and substituting $u = x+4$ back in gives us:

$$y = \frac{1}{6}(x+4)^6 + c$$

You can differentiate using the chain rule to verify that this works.

Example 8.2: Lets evaluate the integral

$$y = \int \cos(3x+4) \, dx$$

Let's use $u = 3x+4$ as a substitution. Lets rearrange for x, differentiate in terms of u and find an expression for dx:

$$u = 3x+4 \implies x = \frac{1}{3}(u-4) \implies \frac{dx}{du} = \frac{1}{3} \implies dx = \frac{1}{3}du$$

Use these to make the substitution:

$$\int \cos(3x+4)\,dx \implies \frac{1}{3}\int \cos(u)\,du$$

Now we can evaluate using the basic laws of integration:

$$\frac{1}{3}\int \cos(u)\,du = \frac{1}{3}\sin(u)+c$$

and substituting $u = 3x+4$ back in gives us:

$$y = \frac{1}{3}\sin(3x+4)+c$$

You can differentiate using the chain rule to verify that this works.

So we see that we can integrate more complicated functions using substitution. We want to use a substitution so that the integration becomes something we can use our table of standard integrals for. However, this is integration by substitution for **indefinite integrals**. In many cases, we want to find the **area under a curve** so the integral has limits. For example,

$$y = \int_{-1}^{1} (3x+2)^4 dx$$

therefore we have to perform **definite integration**. This means that we calculate the integral (using integration by substitution) then evaluate the integral but for the new limits calculated under the substitution.

Let's show how definite **integration by substitution** works in full:

1. Write out the integral in its simplest form:

$$y = \int_{-1}^{1} (3x+2)^4 dx$$

2. Find a new variable (u) as a function of the old variable (x) as a substitution:

$$u = 3x+2$$

3. Rearrange to get an expression for the old variable (x) in terms of the new variable (u):

$$x = \frac{1}{3}(u-2)$$

4. Differentiate with respect to the new variable u:

$$\frac{dx}{du} = \frac{1}{3}$$

5. Get an expression for dx:

$$dx = \frac{du}{3}$$

6. Find the new limits of the integral:

$$\text{When } x = -1, \;\; u = 3 \times (-1) + 2 = -1$$
$$\text{When } x = 1, \;\; u = 3 \times 1 + 2 = 5$$

7. Make the substitution in the original integral

$$y = \int_{-1}^{1} (3x+2)^4 dx \implies y = \int_{-1}^{5} u^4 \frac{du}{3} \implies y = \frac{1}{3}\int_{-1}^{5} u^4 \, du$$

8. Integrate as usual:

$$\begin{aligned} \frac{1}{3}\int_{-1}^{5} u^4 \, du &= \frac{1}{3}\left[\frac{1}{5}u^5\right]_{-1}^{5} \\ &= \frac{1}{3} \times \frac{1}{5} \times (5^5 - 1^5) \\ &= \frac{3124}{15} = 208.267 \end{aligned}$$

Let's see some more examples of this.

Example 8.3: Lets evaluate the integral

$$y = \int_{-4}^{6} (x+4)^5 \, dx$$

Let's use $u = x+4$ as a substitution. Lets rearrange for x, differentiate in terms of u and find an expression for dx:

$$u = x+4 \implies x = u-4 \implies \frac{dx}{du} = 1 \implies dx = du$$

Find the new limits:

$$\text{When } x = -4, \;\; u = -4+4 = 0$$
$$\text{When } x = 6, \;\; u = 6+4 = 10$$

Use these to make the substitution:

$$\int_{-4}^{6} (x+4)^5 \, dx \implies \int_{0}^{10} u^5 du$$

Now we can evaluate using the basic laws of integration:

$$\begin{aligned}\int_0^{10} u^5\,du &= \left[\frac{1}{6}u^6\right]_0^{10} \\ &= \frac{1}{6}\times(10^6-0^6) \\ &= \frac{1,000,000}{6} = 166,666.67\end{aligned}$$

Example 8.4: Lets evaluate the integral

$$y=\int_{-9}^{1}\frac{1}{x+10}dx$$

Let's use $u=x+10$ as a substitution. Lets rearrange for x, differentiate in terms of u and find an expression for dx:

$$u=x+10 \implies x=u-10 \implies \frac{dx}{du}=1 \implies dx=du$$

Find the new limits:

$$\begin{aligned}&\text{When } x=-9,\ \ u=-9+10=1\\ &\text{When } x=1,\ \ u=1+10=11\end{aligned}$$

Use these to make the substitution:

$$\int_{-9}^{1}\frac{1}{x+10}dx \implies \int_1^{11}\frac{1}{u}du$$

Now we can evaluate using the basic laws of integration:

$$\begin{aligned}\int_1^{11}\frac{1}{u}du &= [\ln(u)]_1^{11} \\ &= \ln(11)-\ln(1) \\ &= \ln(11)=2.398\end{aligned}$$